

Tutorial 4 - SPM (wk 9 -Done)

Question 1

Table 1 shows the size oriented metrics of Project team A & B.

Table 1: size oriented metric of Project Team A and Team B

	Non normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programming languages used	Java	Php	Java	Php
Total Lines of Code (LOC)	306,000	256,000		
Errors	288	233		
Defects	30	20		
Pages of Documentation	205	180		

Normalised the metrics of project team A and project team B with KLOC (thousand lines of code). Using the normalised metrics, compare the *quality* of both projects. Provide comments with an analysis of both projects.

Note:

- Provide your answers for the last 2 columns of Table 1.
- Show your workings, round up your answer to 4 decimal points.

Team1

	Non normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programming languages used	Java	Php	Java	Php
Total Lines of Code (LOC)	306,000	256,000	$306,000 / 1000$ $= 306 \text{ KLOC}$	$256,000 / 1000$ $= 256 \text{ KLOC}$
Errors	288	233	$288 / 306$ $= 0.9412$ Errors / KLOC	$233 / 256$ $= 0.9102$ Error / KLOC
Defects	30	20	$30 / 306$ $= 0.098$ defects / KLOC	$20 / 256$ $= 0.0781$ defects / KLOC
Pages of Documentation	205	180	$205 / 306$ $= 0.6699$ Pages / KLOC	$180 / 256$ $= 0.7031$ Pages / KLOC

Comments:

TeamA has > errors than TeamB (0.9412 vs 0.9102)

TeamA has > defects than TeamB (0.0980 vs 0.0781)

TeamB has > pages of documentation than TeamA (0.7031 vs 0.6699)

Conclusion

Project B have higher quality based on:

Higher Correctness (Errors):

- Lower error per KLOC as compared to Project A
- Lower defects per KLOC as compared to Project A

Higher Maintainability (Documentation):

- Higher Page of Documentation, allowing the maintenance staff to maintain it easily, therefore it has higher quality of maintainability

Team 3

	Non normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programmin g languages used	Java	PHP	Java	PHP
Total Lines of Code(LOC)	306,000	256,000	306000/1000= 306 KLOC	256000/1000 =256 KLOC
Errors	288	233	288/306 =0.9412 error/KLOC	233/256 =0.9102 error/KLOC
Defects	30	20	30/306 =0.0980 defect/KLOC	20/256 =0.0781 defect/KLO C
Pages of Documentati on	205	180	205/306 =0.6699 pages/KLOC	180/256 =0.7031 pages/KLOC

comments:

Team B's project shows better quality and reliability than Team A's. It has fewer errors and defects per 1000 LOC and more documentation pages, making it more reliable and maintainable compared to Team A's project.

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Normalised the metrics of project team A and project team B with KLOC (thousand lines of code). Using the normalised metrics, compare the *quality* of both projects. Provide comments with an analysis of both projects.

Note:

-Provide your answers for the last 2 columns of Table 1.

- Show your workings, round up your answer to 4 decimal points.

Team 4

	Non Normalized		Normalized KLOC	
	Team A	Team B	Team A	Team B
Pro Languages	Java	PHP	Java	PHP
Total lines of code (LOC)	306,000	256,000	$306000/1000$ =306 KLOC	$256000/1000$ =256 KLOC
Errors	288	233	$288/306$ =0.9412 error /KLOC	$233/256$ =0.9102 error /KLOC
Defects	30	20	$30/306$ =0.0980 defects / KLOC	$20 / 256$ =0.0781 defects / KLOC
Pages of Documentation	205	180	$205 / 306$ = 0.6699 pages/KLOC	$180/256$ =0.7031 pages/KLOC

Comment:

Correctness(Errors)

On team B's project, it has less errors made compared to Team A ($0.9102 < 0.9412$). We can say that Team B has higher correctness due to low errors.

Maintainability(Pages of Documentation)

Team B has higher pages of documentation, compared to Team A ($0.7031 > 0.6699$)

Team B has a higher quality of maintainability.

Conclusion:

Team B's project has a higher quality.

Team 3

Question 1

Table 1 shows the size oriented metrics of Project team A & B.

Table 1: size oriented metric of Project Team A and Team B

	Non normalised		Normalised with KLOC	
	Team A	Team B	Team A	Team B
Programming languages used	Java	Php	Java	Php
Total Lines of Code (LOC)	306,000	256,000	306	256
Errors	288	233	0.9412	0.9102
Defects	30	20	0.0980	0.0781
Pages of Documentation	205	180	0.6699	0.7031

Normalised the metrics of project team A and project team B with KLOC (thousand lines of code). Using the normalised metrics, compare the *quality* of both projects. Provide comments with an analysis of both projects.

Note:

- Provide your answers for the last 2 columns of Table 1.
- Show your workings, round up your answer to 4 decimal points.

$$KLOC = LOC / 1000$$

$$\text{Team A} = 306000 / 1000 = 306 \text{ KLOC}$$

$$B = 256000 / 1000 = 256 \text{ KLOC}$$

$$\text{Defect per KLOC}$$

$$A = 30 / 306 = 0.098$$

$$B = 20 / 256 = 0.0781$$

$$\text{Error per KLOC}$$

$$A = 288 / 306 = 0.9412$$

$$B = 233 / 256 = 0.9102$$

$$\text{Pages of documentation / KLOC}$$

$$A = 205 / 306 = 0.6699$$

$$B = 180 / 256 = 0.7031$$

∴ Team B is better because lower errors and defect, higher documentation

Conclusion

- project b has lower errors than project a which is 0.9102 vs 0.9412. Project b has better product quality.
- Project b has more documentation per KLOC than project a which is 0.7031 vs 0.6699. It means that the project b is support maintainability.
- Therefore, project b's quality is higher than project a

Question 2

Team 3

Question 2

For a) and b), show the workings in your calculation. Round up your answer to two decimal places.

a) Referring to Table 1, calculate the *function points (FP)* for the both projects.

FP formula: Count Total $\times [0.65 + 0.01 \sum Fi]$

Table 1: Information domain values of Project A and B

Measurement parameters	Weighting factor	Project A's Count	Project B's Count
Number of inputs	Complex (5)	4	4
Number of outputs	Simple (2)	6	5
Number of inquiries	Average (4)	8	8
Number of files	Average (3)	4	2
Number of external interfaces	Simple (2)	2	1
The adjustment values $\sum Fi$ for both projects		50	40

Table 2: Defects found and documentation of Project A and B

	Project A's Count	Project B's Count
Defects	90	70
Pages of documentation	120	130

b) Referring to Table 2, calculate the normalised defects and pages of documentation based on the calculated answer in part a) function point. Compare which project has *higher maintainability* based on the calculated result. Provide comments with an analysis of both projects.

a. Project A

$$\begin{aligned} FP(A) &= \text{count total} \times [0.65 + 0.01 \sum Fi] \\ &= (4 \times 5 + 6 \times 2 + 8 \times 4 + 4 \times 3 + 2 \times 2) \times [0.65 + 0.01 \times 50] \\ &= 80 \times 1.15 \\ &= 92 \end{aligned}$$

Project B

$$\begin{aligned} FP(B) &= \text{count total} \times [0.65 + 0.01 \sum Fi] \\ &= (4 \times 5 + 5 \times 2 + 8 \times 4 + 2 \times 3 + 1 \times 2) \times [0.65 + 0.01 \times 40] \\ &= 70 \times 1.05 \\ &= 73.50 \end{aligned}$$

b. Defects/FP

$$\begin{aligned} A &= 90/92 = 0.98 \\ B &= 70/73.50 = 0.95 \end{aligned}$$

Pages of documentation /FP

$$\begin{aligned} A &= 120/92 = 1.3 \\ B &= 130/73.50 = 1.77 \end{aligned}$$

$\therefore$ Project B has higher maintainability based on calculated results. It is because project B is good quality (0.95) and it is higher maintainability (1.77)

Conclusion

Project B has higher quality due to:

Correctness

- Lower Defects / FP, lesser defects.
- Higher degree of correctness

Maintainability

- Project B has a higher number Pages of Doc/FP than Project A. This means that there is more info about P(B). Hence, P(B) has a higher degree of maintainability.

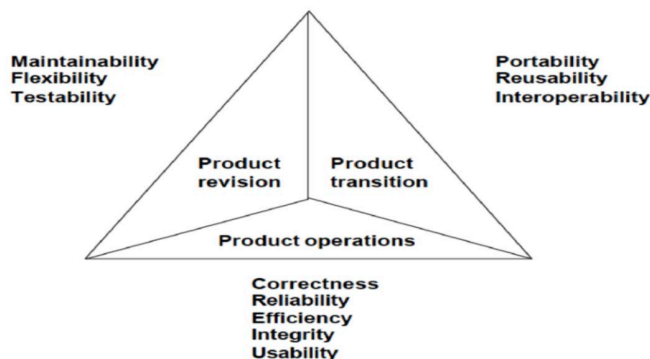
Question 3

Question 3

McCall's Software Quality Factor model is used to evaluate the quality of software products.

a) Describe the following categories

- Product Operation, Product Revision and Product Transition.



Refer to Appendix 4.1,
software quality attributes in
each categories

b) Refer to the case study in Tutorial 1 Question 2 (T-shirt2U company).

Explain ONE (1) quality attribute applicable to the T-shirt2U's new ordering system and provide ONE(1) metric for measuring it.

Product Operation

This category evaluates how well the software performs during the system execution. It focuses on attributes such as:

- **Correctness** – whether the software produces the expected results.
- **Reliability** – ability to operate without failure for a given time.
- **Efficiency** – use of system resources (CPU, memory, response time).
- **Integrity** – protection from unauthorized access (security).
- **Usability** – ease of use and user-friendliness.

Product Revision

This category is about how easy it is to make changes to the software after delivery. It emphasizes:

- **Maintainability** – effort needed to locate and fix errors.
- **Flexibility** – effort needed to adapt the software to new requirements.
- **Testability** – effort needed to verify modifications or improvements.

Product Transition

This category concerns how easily the software can adapt to a new environment or be reused. It includes:

- **Portability** – ability to run on different hardware/software platforms.
- **Reusability** – use of software components in other applications.
- **Interoperability** – ability to interact and exchange data with other systems.

B) Reliability (Product Operation)

Explanation: For T-shirt2U's online ordering system, reliability is crucial because customers need the system to be consistently available when browsing products, adding items to carts, making online payments, and tracking fulfillment. Any downtime or failure could result in lost sales and prevent the company from achieving its business goal of increasing annual sales by RM50,000.

Metric: Mean Time Between Failures (MTBF)

- This metric measures the average time the system operates without a failure.
- A higher MTBF indicates greater reliability (e.g., aiming for 500+ hours between failures).

Alternative:

Lower number of errors and defects per function point

Question 4 (Team 4 (Team DNM) - Low Jian Hao present)

Question 4

Identify and explain TWO (2) characteristics of good metrics to measure the quality attributes of the system.

1. Programming language independent

Metrics should be based on the analysis model, the design model, or the structure of the program itself. LOC is programming language dependent. (eg Assembly vs C++)

2. Simple and computable

Relatively easy to learn how to derive the metric. Its computation should NOT demand excessive amounts of effort or time.

3. Consistent in its use of units and dimensions

The mathematical computation of the metric should use measures that do not lead to bizarre combinations of units. E.g. use size and/or FP throughout all projects

Refer to Chp5 Slide 47 & 48 for other points